

Normalization:

All 8 tables were normalized manually to BCNF. The process for each table is below.

6.1 Users Table

Attributes: user-id, name, email, password, role, created-at

- 1NF: All values are atomic; no repeating groups
- 2NF: Primary key is single-attribute (user-id) - no partial dependency possible
- 3NF: No non-key attribute transitively depends on user-id via another non-key
- BCNF: Two candidate keys exist (user-id, email) both are superkeys; every determinant is a candidate key.

6.2 Categories table

Attributes: category-id, category-name

- FD: category-id $\rightarrow$ category-name
- BCNF: only one nontrivial FD exists; determinant (category-id) is a candidate key.

6.3 Product Table

Attributes: product-id, category-id, name, price, stock, description, image-path

- original issue: category-name would create a transitive dependency (product-id $\rightarrow$ category-id $\rightarrow$ category-name)
- solution: category details separated into the categories table - transitive dependency eliminated.
- BCNF: product-id is the sole candidate key; all the attributes depends directly and only on it.

6.4 Orders Table

Attributes: order-id, user-id, order-date, total-amount, status, phone-number, delivery-address

- FD: order-id $\rightarrow$ all other attributes
- BCNF: order-id is the only candidate key no partial or transitive dependencies

6.5 OrderItems Table

Attributes: item-id, order-id, product-id, quantity, unit-price

- Designate: unit-price is stored as a snapshot of the price at time of order - this prevents historical data loss if a product price changes later.
- FD: item-id $\rightarrow$ order-id, product-id, quantity, unit-price
- BCNF: item-id is the sole candidate key; no anomalies

6.6 Cart Table

Attributes: cart-id, user-id, product-id, quantity

- FD: cart-id $\rightarrow$ user-id, product-id, quantity
- Composite uniqueness: (user-id, product-id) pair is logically unique; on duplicate add, quantity is incremented via UPDATE rather than INSERT
- BCN ✓

6.7 Wishlist Table

Attributes: wishlist-id, user-id, product-id

- Duplicate prevention enforced in application logic before INSERT
- BCNF ✓

6.8 Payment Table

Attributes: payment-id, order-id, amount, payment-date, method

- FD: payment-id $\rightarrow$ order-id, amount, payment-date, method
- order-id is a unique FK (1:1 with orders)
- BCNF ✓